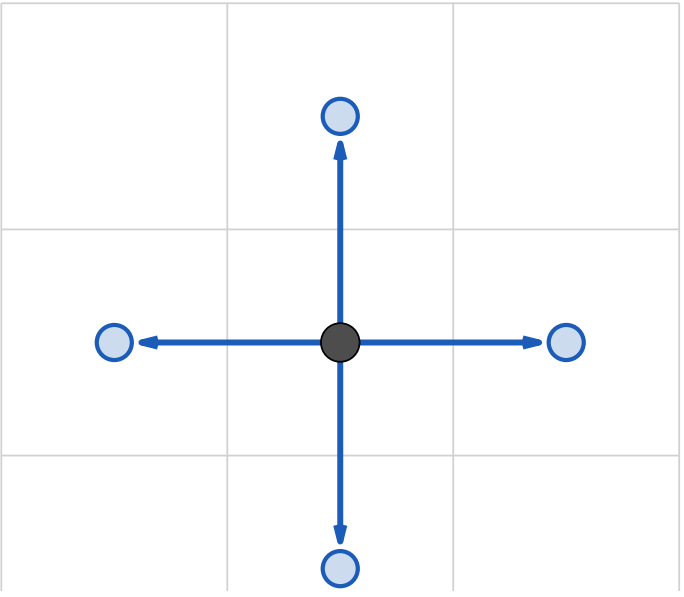
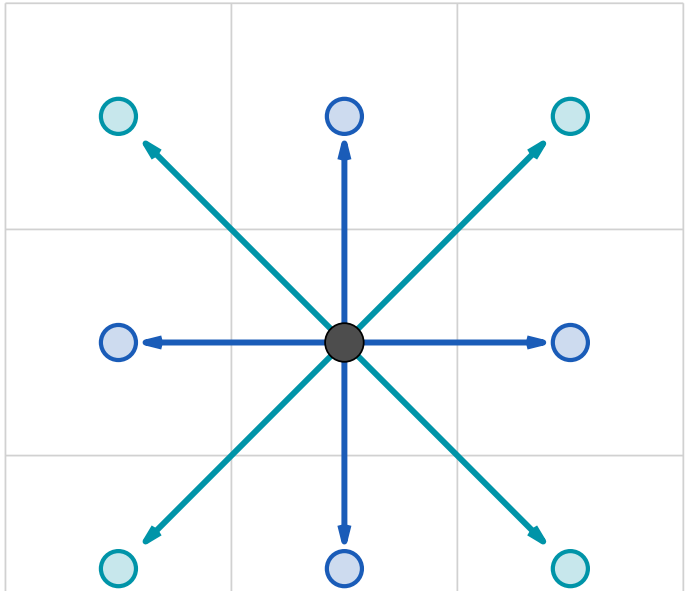
Edge cost: 1



(a) Four-connected

Cardinal: 1; diagonal: $\sqrt{2}$



(b) Eight-connected